Ask the Reviews: Grounded, Constraint-Aware Question Answering over 61,167 Airline Passenger Reviews

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Abstract

We address question answering over passenger reviews for travellers choosing an airline. Real questions are constrained (a route, a cabin, a traveller type), they ask about fine-grained attributes such as legroom or seat cleanliness that structured ratings do not cover, and they need a verdict the user can check. We build a retrieval-augmented system over 61,167 Skytrax reviews split into 546,139 sentence passages. The query is parsed into hard metadata filters and one of 46 sub-attributes; evidence is retrieved with BM25 and MiniLM dense search fused by reciprocal rank fusion; an offline sentence classifier labels subjectivity, polarity and aspect so that the verdict is a count over *every* matching review; and a local Qwen3-4B model writes a cited answer that program guards accept only if its citations, numbers and direction agree with the evidence. On 30 airline × sub-attribute queries the hybrid retriever reaches NDCG@10 0.810 and MRR 0.961, while a general-purpose cross-encoder *lowers* NDCG@10 by 0.066 (95% CI $[-0.114, -0.019]$). Replacing the rule-based classifier with a linear student trained on weak labels plus labels distilled from an LLM raises three-class macro-F1 on our 1,000-sentence human-labelled set from 57.2 to 78.5 and joint aspect–polarity accuracy by 7.4 points (CI $[+4.5, +10.4]$), while relabelling the whole corpus in 18 s. On 20 end-to-end questions the guarded system scores 0.84 faithfulness and 0.99 answer relevance, with 16 of the 20 final answers written by the program fallback. A full factorial ablation of the three answer-side components (160 answers) finds balanced evidence plus validation the most faithful configuration (+0.053, CI $[+0.004, +0.106]$) and shows that their gains are not additive. Text-derived airline rankings agree with held-out passenger ratings (Spearman $\rho = 0.81$ over 58 airlines). Everything runs locally at zero API cost.

1 Introduction

A traveller deciding between airlines does not ask "what do reviews say about seats?". They ask "flying Singapore to London in economy, which airline has the best legroom?", or "how clean are Singapore Airlines seats?". Such questions combine hard constraints with an attribute that is usually mentioned only in free text, and the answer is only useful if the user can see how much evidence lies behind it and check it. Review sites hold this evidence, but in tens of thousands of long reviews that no one can read before booking.

We frame the task as retrieval-augmented opinion question answering, and derive three requirements that shape the whole design:

R1 **Constraints must hold.** Route, cabin, traveller type and date are filters on the evidence, not hints to a ranker.

R2 **Fine-grained attributes need text.** The data rates only six coarse aspects; legroom, recline or meal portion size appear only in the review text.

R3 **The verdict must be traceable.** The overall direction is computed over all matching reviews, and every claim in an answer cites a review sentence.

The contributions of this report are:

- A cleaned, sentence-level knowledge base with structured metadata, and an audit of the data problems that would otherwise damage retrieval (Section 3).
- A retrieval pipeline that turns constraints into filters and routes 46 sub-attributes with query expansion, evaluated with rank-aware metrics and bootstrap intervals, including a negative result for cross-encoder reranking (Sections 4 and 5.2).
- A distillation protocol that moves an LLM labeller into a fast classifier with pre-registered selection that never reads the gold set (Section 5.5).
- Three answer-side components (full-pool statistics, balanced and label-consistent evidence, validation with repair and fallback) evaluated in a full 2^3 factorial ablation, and an airline-ranking validation against held-out passenger ratings (Sections 5.7 and 5.8).

The brief poses five questions; Table 1 shows where each is answered.

Table 1: Where each question of the assignment brief is answered.

Q	Topic	Sections
Q1	Corpus gathering, cleaning, chunking, storage, application, statistics	§3
Q2	User interface; five queries and their speed	§4.4, §5.3
Q3	Retrieval innovations; rank-aware evaluation	§4.1, §5.2
Q4	Approach, evaluation dataset, classification and RAG metrics, performance	§4.2, §5.1, §5.4, §5.6, §5.9
Q5	Downstream innovations; ablation studies	§4.3, §5.5, §5.7, §5.8

2 Related Work

Retrieval. Sparse retrieval with BM25 [1] remains a strong baseline; dense bi-encoders such as Sentence-BERT [2] with MiniLM [3] capture paraphrases and are searched efficiently with FAISS [4]. Reciprocal rank fusion [5] combines rankings without calibrating scores, and cross-encoders trained on MS MARCO [6, 7] are the usual second-stage rerankers. BEIR [8] showed that BM25 stays hard to beat out of domain, which E1 confirms. Larger encoders such as BGE [9] score higher on embedding benchmarks, and HyDE [10] expands a query with an LLM-written hypothetical answer. We keep MiniLM because 529,031 sentences must be encoded on a laptop, and expand queries from a fixed taxonomy instead of an LLM: the expansion is inspectable and adds no model call per query. We evaluate with NDCG [11] and related rank-aware metrics.

Aspect-based sentiment. Aspect-based sentiment analysis [12, 13] has moved from lexicon and rule systems such as SenticNet [14] to fine-tuned transformers [15] and instruction-tuned generators such as InstructABSA [16], which report state-of-the-art results on SemEval benchmarks. Both need in-domain aspect labels, which our eight-aspect scheme does not have beyond the gold set. A systematic study of LLMs for sentiment [17] found them strong zero-shot on simple polarity but behind fine-tuned small models on structured tasks such as ABSA, and strongest when labels are scarce. LLM labels can replace part of the human annotation effort [18], and knowledge distillation [19] transfers an expensive model's behaviour into a cheaper one; zero-shot NLI [20, 21] is another label-free classifier. Our design follows from these results: a zero-shot LLM teacher, where labels are scarce, distilled into a fast student for the whole corpus (Section 4.2), with NLI as a second opinion on objectivity.

Grounded generation. Retrieval-augmented generation [22] conditions a generator on retrieved passages; RAGAS [23] defines reference-free answer metrics (faithfulness, answer relevance, context precision) that we adopt. ALCE [24] evaluates answers that cite their sources, and Self-RAG [25] trains the generator to critique its own retrieval and output. Training a generator is out of reach for a laptop 4B model, so we enforce grounding outside it: citations must point to displayed evidence, numbers must match program-computed counts, and a program-written answer replaces a draft that fails (Section 4.3).

3 Knowledge Base Construction

Source. The corpus is the public Skytrax review collection released on Kaggle (`efehandanisman/skytrax-airline-reviews`) [26]. Each record holds the review text with airline, cabin, traveller type, route, date flown, aircraft, six 1–5 aspect ratings (seat comfort, cabin service, food and beverages, entertainment, ground service, value for money), a 1–10 overall score and a recommend flag. We download it with `kagglehub`; no scraping is involved.

Why this domain. The brief asks for enough data and a balanced task. Opinions are mixed within and across airlines (40% of opinionated gold sentences are positive, 60% negative), and the ratings give free weak supervision. We rejected paper peer reviews (too hedged for 80% inter-annotator agreement), MOOC reviews (overwhelmingly five-star), cryptocurrency (the brief's own example, swinging with the market) and Steam reviews (the most likely to overlap with other groups).

Cleaning. We audited all 65,947 non-empty reviews before writing cleaning code; Table 2 lists what we found. Two problems affected over 40% of reviews. We deliberately did *not* lower-case, strip punctuation or correct spelling: the text is shown to users as evidence, and exclamation marks and intensifiers are sentiment signals. Lower-casing happens only inside the BM25 index.

Chunking. We split every review into sentences with spaCy [28]. A sentence usually carries one opinion about one aspect, so a retrieval hit, a classifier label and a citation all refer to the same span; longer windows mix aspects, because a typical review moves from check-in to seat to food (Fig. 1, bottom). Each sentence keeps its `review_id` and the full review, so the interface can expand a hit to its parent (parent-document retrieval). Sentences of fewer than three words or five letters are dropped at split time; 16,772 route-only lines and 336 non-English sentences are excluded from retrieval (19 sentence fragments are flagged but kept), leaving a pool of 529,031 of the 546,139 sentences.

Storage. The result is one table, `data/processed/chunks.parquet` (31 columns: sentence, parent review, metadata, ratings, route parsed into origin, destination and stop-over, quality flags), which is the single data contract of all three subgroups. zstd compression reduces it from 96 MB to 31 MB. On it we build a BM25 index (6 s), a FAISS `IndexFlatIP` over L2-normalised 384-d `all-MiniLM-L6-v2` embeddings (4.6 min, mostly encoding) and the label index of Section 4.2. A full rebuild takes about 14 minutes.

Table 2: Data problems found in the raw corpus and how they were fixed (`src/clean_text.py`).

Problem	Share	Example and fix
Blank spreadsheet rows	50.0% of rows	131,895 rows, of which 65,947 carry a review; the rest are dropped.
Mojibake (UTF-8 read as cp1252)	42.4%	`wasnâ€™t` → `wasn't`; `DÃ¼sseldorf` → `Düsseldorf`. Re-encoding whole reviews left 2.14% broken, repairing suspicious spans 0.37%; `ftfy` [27] plus a span-level second pass leaves 6 flagged sentences (0.001%), all genuine Spanish, French or Portuguese.
Site boilerplate	43.5%	Prefixes `Trip Verified |` (13,670), `Verified Review |` (12,126), `Not Verified |` (1,714) and a trailing `···Ranking`, removed with anchored patterns; otherwise every query on "verified" would match them.
Duplicate reviews	7.2%	Exact de-duplication: 65,947 → 61,172 unique texts.
Too-short reviews	<0.01%	Under 20 characters after cleaning: 61,172 → 61,167.

Table 3: Corpus statistics after cleaning and chunking. Words are whitespace tokens; types are distinct lower-cased tokens.

Documents (reviews)	61,167
Chunks (sentences)	546,139
of which retrievable	529,031
Words (tokens)	8,276,082
Types (unique words)	130,385
Airlines	81
Review dates	Aug 2002 – Jun 2019
Reviews by cabin: Economy	45,163
Business / Premium / First	9,589 / 2,411 / 1,532
Cabin not given	2,472

Statistics. Table 3 gives the size of the corpus, well above the required 10,000 documents and 100,000 words. Its main bias is towards economy class (74% of reviews) and towards large airlines (Spirit, American, United and British Airways have about 2,800 reviews each); answers therefore state the number of matching reviews so users can judge how thin the evidence is.

Application and sample queries. Users are travellers choosing an airline and analysts comparing service quality. They ask in English or Chinese, or set filters directly, and receive a cited verdict, a positive/negative tally over all matching reviews, the rating summary and the evidence. Typical queries: *How is the legroom in Singapore Airlines economy?* (one sub-attribute); *Flying Singapore to London in economy, which airline has the best legroom?* (route constraint, open comparison); *Compare Qatar Airways and British Airways business class seat recline* (two airlines); *Singapore Airlines economy: food, value for money and ground service?* (several aspects); and the Chinese *阿联酋航空经济舱的退款和改签处理怎么样?* (Emirates refunds and rebooking).

4 Proposed System

Figure 2 shows the pipeline. Indexes and labels are built offline; per query, filters are applied before retrieval, and the generator's draft is accepted only if program checks pass. The system runs on an Apple M1 Pro laptop with 16 GB of memory; no hosted RAG service or paid API is used.

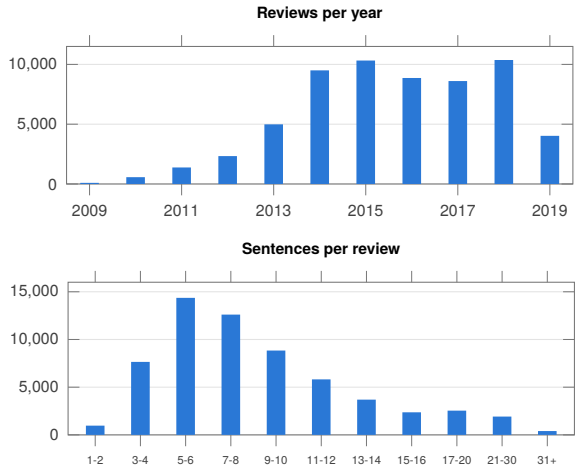


Figure 1: Top: review volume grows from 2013 (115 reviews before 2009 omitted); 2019 is partial. Bottom: most reviews have 5–10 sentences and therefore mix several aspects, which motivates sentence-level passages.

4.1 Retrieval (Q3)

Baseline. The first version followed the course design: BM25 (`rank-bm25`) over the retrievable sentences; exact inner-product search over normalised MiniLM embeddings; and reciprocal rank fusion of the two top-100 lists, $\mathrm{RRF}(d) = \sum_r 1/(k + \mathrm{rank}_r(d))$ with $k = 60$, which fuses ranks because BM25 scores and cosine similarities live on different scales. An optional cross-encoder (`ms-marco-MiniLM-L-6-v2`) re-scores the fused top-100. One early bug is worth recording: duplicate sentences tied in score and the two retrievers broke ties differently, so fusion split their votes across different `chunk_ids` of the same text. Both retrievers now sort by (score, row) and de-duplicate by text before fusion.

Innovations. The baseline handled generic queries such as "rude cabin crew" well but failed on the questions users actually ask. Table 4 lists each failure and the change that fixed it. Hard constraints (R1) are metadata and timeline search: a route or date-window question must not be answered from other routes or years. Sub-attribute routing (R2) is query expansion driven by a taxonomy of 46 sub-attributes (Appendix E) rather than by pseudo-relevance feedback, which would drift towards the dominant aspect of the top documents. The full-slice scan (R3) makes counts

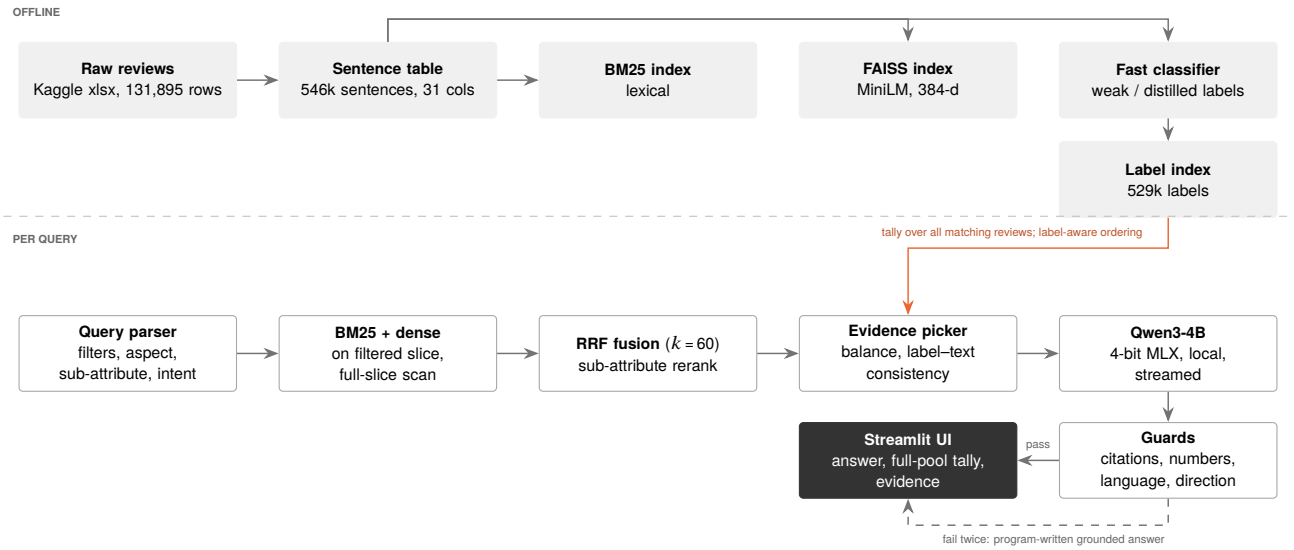


Figure 2: System architecture. Offline, one sentence table feeds three indexes. Per query, structured filters are applied before lexical and dense retrieval; the generator writes only from selected evidence and program-computed counts, and its draft is shown only if program checks pass (one automatic repair, then a program-written fallback).

such as "95 direct legroom sentences in 86 reviews" exact instead of dependent on k, which is what allows the verdict to be computed over all evidence. E1b (Table 8) measures each change.

4.2 Sentence classification (Q4)

Every sentence receives three labels: subjectivity, polarity (for subjective sentences) and one of eight aspects (seat, staff, food, entertainment, ground, value, overall, other). The brief requires subjectivity then polarity detection; adding the aspect makes the task ABSA [12], which attribute-level questions need. Because the answer must count opinions over all matching reviews, labels must exist before a query arrives, so we combine three components (Table 5): an accurate but slow LLM teacher, a fast student that labels the corpus, and a grounded generator. The constraint parser, the taxonomy and the verification rules form the symbolic half of this hybrid symbolic–subsymbolic design.

Weak supervision. A sentence that mentions exactly one aspect, according to keyword rules, inherits that aspect's 1–5 rating (4–5 positive, 1–2 negative, 3 dropped). This yields 231,779 training and 58,220 development sentences, split by parent review, with all gold sentences, their reviews and identical texts excluded (overlap verified to be zero). Weak labels are noisy and contain only 221 objective examples (179 in the training split), which is why weak-only models rarely predict "objective".

Preprocessing. The cleaning of Section 3 matters as much for classification as for retrieval: broken tokens such as `donâ€™t` hide negations. We did not apply microtext normalisation: the reviews are long-form prose, and the audit found encoding damage, not informal spelling, to be the problem. Case and punctuation are kept for the TF-IDF features.

4.3 Grounded generation (Q5)

Each answer-side component responds to an observed failure. For the ablation in Section 5.7 we group them into three factors: **X**, full-pool statistics; **Y**, balanced and label-consistent examples; and **Z**, validation with one repair and a program-written fallback.

X: full-pool verdict. Early on, the direction came from the ten sentences on screen; for one airline's legroom these were mostly negative, so the answer called legroom poor although 79.6% of the 98 directional matching reviews were positive. The verdict is now a count over every review that directly matches the sub-attribute, one vote per review; displayed sentences are examples and cannot change it. An explicit positive or negative verdict requires at least ten decidable reviews and a 95% Wilson interval entirely above or below 50%; otherwise the answer says the evidence is insufficient or mixed.

Y: balanced evidence. The answer cites at least one mainstream and one counter-example sentence when both exist, so the minority view is visible and the generator cannot present a one-sided picture.

Y: label–text consistency filter. Some sentences carry a label their text plainly contradicts. The evidence picker counts positive and negative cue words ("rude", "cramped", "delayed" vs. "friendly", "plentiful", "on time"); when they outweigh the label in the opposite direction, the sentence is skipped as a representative example (it still counts in the tally).

Z: grounding guards. A validator checks that every citation refers to displayed evidence, every number matches the program-computed counts, the language matches the interface, and the stated direction agrees with the full-pool verdict. A failing draft gets one repair with the errors listed;

Table 4: Queries that failed in the first version, and the retrieval change that fixed each.

Query	First version	Now, and why
"Singapore to London in economy, which airline has the best legroom?"	Route words treated as keywords; results mixed routes and cabins, and the ranking ignored legroom.	**Structured hard constraints.** Origin, destination and cabin become filters (LHR, Heathrow, LGW, Gatwick map to one destination set); three candidates are chosen by rating and sample size, then legroom is retrieved for each.
"How clean are Singapore Airlines seats?"	Fell back to the seat-comfort rating and generic comfort sentences.	**Sub-attribute routing with query expansion.** Seat cleanliness is one of 46 sub-attributes with expansion terms; answers now cite crumbs, coffee stains and dirty tray tables.
"新航经济舱的腿部空间好不好?" (Chinese)	Unsupported: corpus and encoder are English-only.	**Cross-lingual querying.** Chinese airline, cabin and attribute names map to canonical values and English expansion terms; evidence is shown translated with the original kept.
"What food do they serve on Singapore Airlines?"	Answered with the food rating and a positive/negative split.	**Intent routing.** Fact-type questions list cited instances (satay, rice, desserts) instead of a sentiment summary.
Any sub-attribute question on a large airline	Top-k saw only the most similar sentences, so direct mentions were missed and the verdict depended on k.	**Full-slice scan with label-aware ordering.** Every sentence in the filtered slice that names the sub-attribute is collected and reranked; hybrid retrieval adds paraphrases; labels reorder but never delete candidates.

Table 5: The three model components and the reason for each.

Component	What it is	Why
LLM teacher (r36)	Qwen3-4B-Instruct-2507 [29], 4-bit MLX; r36 is the best of the labelling configurations we compared. Two-step closed-choice prompts (one of eight aspects, then a polarity). A sentence is objective only if an independent classifier agrees: zero-shot NLI with DeBERTa-v3-base [21], averaged with a MiniLM head trained on weak labels.	Most accurate labeller we have (Table 10), but ≈0.5 s per sentence: about three days for the corpus.
Fast student	TF-IDF (1–2-grams, 20,000 features) concatenated with frozen MiniLM embeddings; one logistic regression per task; trained on weak labels plus teacher labels (Section 5.5).	Labels all 529,031 sentences in 18 s once embeddings are cached, so the label index can be rebuilt whenever the model improves.
Grounded generator	The same Qwen3-4B, served locally with streaming; writes only from selected evidence and program-computed counts.	Fluent answers to multi-constraint questions, no API cost, no data leaving the machine; the guards (Section 4.3) make unsupported claims visible.

if that fails, a program-written answer is shown. In comparisons, a winner is declared only when the rating gap is significant and practically meaningful.

4.4 User interface (Q2)

The interface is a single Streamlit page (Figs. 3 and 4): one question box, filters in a sidebar, results in a fixed order. Each element solves a specific problem. *Sidebar filters* (airline, 2–3 airlines to compare, cabin, aspect, route, date) show what the parser understood and let the user correct it. The *answer* carries numbered citations and states its mode (model draft, repaired, or program-written). The *sentiment overview* gives positive and negative shares over all matching reviews with their counts; this is the verdict. The *rating summary* reports the airline's own ratings under the same filters with a significance check. *Evidence tabs*, one per airline, show each sentence with cabin, sub-attribute match, label and expandable parent review. A *language switch* changes page, answer and evidence together; in Chinese mode, evidence is translated by the local model while the downloadable JSON keeps the original.

5 Experiments

5.1 Setup and evaluation data

E1–E5 and E8 run on an Apple M1 Pro (16 GB); E6 and E7 ran on a Linux machine with an AMD Radeon RX 7800 XT. Intervals are 95% bootstrap intervals at the query, question or parent-review level (10,000 resamples for E1 and E5, 2,000 for E4, E6 and E7). The evaluation data are: 30 airline × sub-attribute retrieval units (E1); 20 fixed end-to-end questions written in Chinese (12 single-airline, 5 comparisons, 3 multi-aspect; E2, E5); a 1,000-sentence human-labelled gold set (E3, E4); 20 English single-attribute ques-

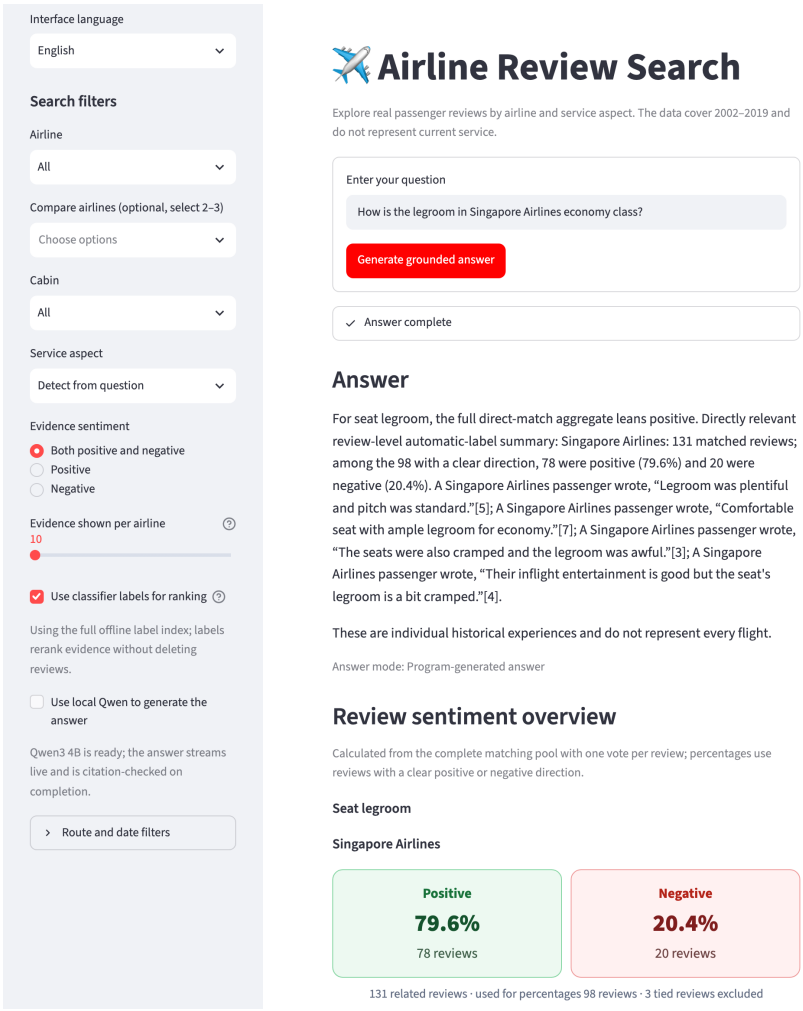


Figure 3: The interface, single-airline question in English mode. The parser detected airline, aspect (seat) and sub-attribute (legroom). The program-written answer cites four sentences; the overview counts 131 matching reviews, 98 of them with a clear direction (79.6% positive). The sidebar holds the structured filters (airline, airlines to compare, cabin, aspect, evidence polarity, and route and date under the last expander).

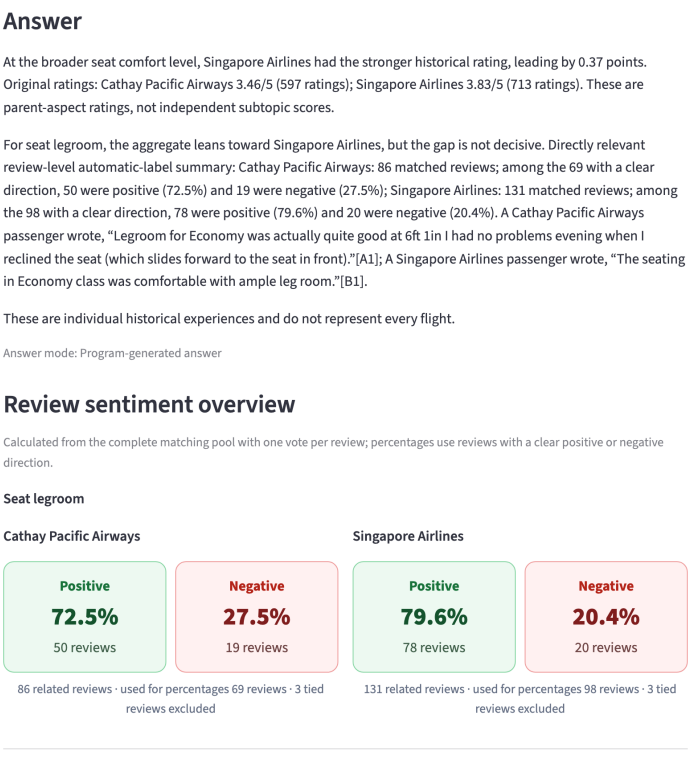
tions for the factorial ablation (E6, Appendix D); and held-out passenger ratings for the airline ranking (E7).

Gold set (Q4). Candidates came from 478,969 sentences after removing route lines, duplicates, sentences under five words and reviews without an overall score. We sampled half from reviews scoring ≥7 and half from ≤4, about 120 per rated aspect, plus sentences with no clear aspect, and shuffled them. Three annotators labelled every sentence independently seeing *only the sentence*: airline and ratings were hidden because ratings are also the weak-label source. A written guideline fixed edge cases such as mixed sentiment. On the frozen 1,000 sentences, agreement exceeds 80% on all three tasks (Table 6); subjectivity has moderate Fleiss' κ [30] because only 8% of sentences are objective. Across all 1,100 labelled sentences, the two tasks the brief requires, subjectivity and polarity, still exceed 80%, but aspect agreement is 75.2%: the 100 sentences left out include 15 that the guideline drops as unclear and 52 on which the three annotators chose three different aspects. Disagreements were resolved by majority vote, then by AI-assisted adjudication confirmed one by one by a human reviewer, who changed 9 sentences. Sampling balanced the review scores (502 vs. 498 sentences) but not the labels: 143 sentences from high-scoring reviews are complaints, while only 49 from low-scoring reviews are praise. We kept this natural skew rather than resample after labelling, and report macro and per-class scores. The set has 548 negative, 371 positive and 81 objective sentences, no duplicate texts, and is stored as `data/eval/eval.xls` in standard benchmark format (Appendix B).

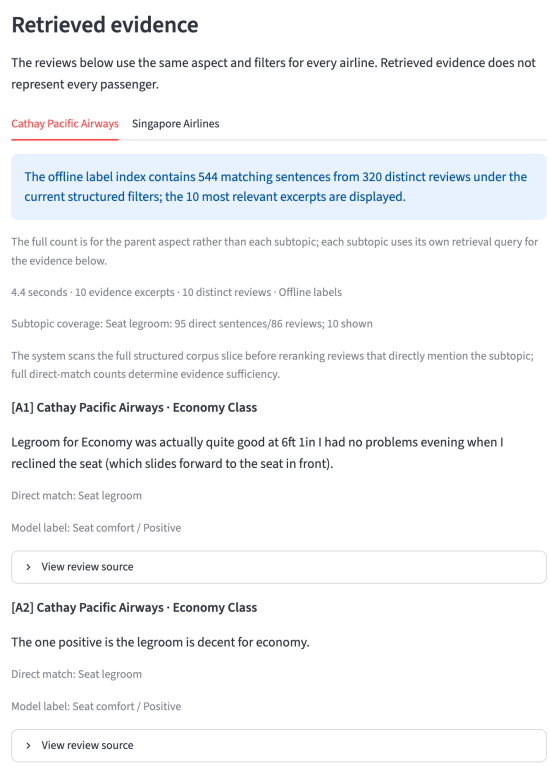
Table 6: Pairwise agreement of the three original annotators. Frozen: the 1,000 gold sentences (3,000 pairs), with Fleiss' κ. All: every labelled sentence (1,100; 3,300 pairs).

Field	Agreeing	Frozen	κ	All
Subjectivity	2,783	92.8%	0.572	90.9%
Polarity (incl. NA)	2,730	91.0%	0.841	89.3%
Aspect (8 classes)	2,426	80.9%	0.778	75.2%
All three at once	2,250	75.0%	0.732	69.8%

Scope of the gold set. We labelled 1,100 sentences and froze 1,000, chosen *after* labelling by keeping sentences whose original aspect labels agreed; agreement on the frozen set is therefore higher than on all labelled sentences (Table 6). Some earlier classifier variants were also scored on these sentences during development. We treat the set as a well-adjudicated benchmark rather than a pristine test set; for the distilled model (E4), selection never read it and it was scored exactly once.



(a) Answer, with the rating test, and full-pool tally.



(b) Evidence tab with coverage counts.

Figure 4: Comparison of Cathay Pacific and Singapore Airlines on economy legroom. Singapore Airlines' higher seat-comfort rating (3.83 vs. 3.46 of 5) is significant and reported for the parent aspect; the legroom tallies (72.5% vs. 79.6% positive) are reported as not decisive. The evidence tab shows 95 direct legroom sentences in 86 reviews and a retrieval time of 4.4 s (a separate run from Table 9).

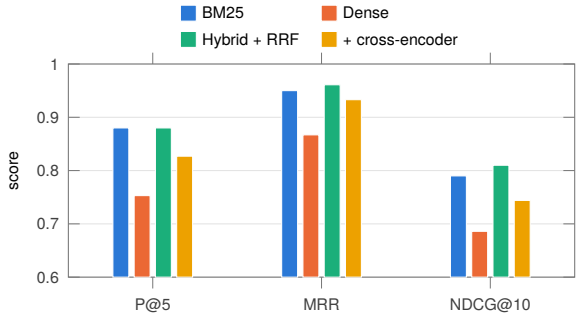


Figure 5: E1: precision-oriented metrics from Table 7 (axis starts at 0.6).

5.2 E1: Does hybrid retrieval rank better? (Q3)

Protocol. Each of the 30 units (e.g. *Singapore Airlines, legroom*) uses the same metadata constraints for every configuration. Relevance is graded automatically: a sentence that explicitly names the sub-attribute is highly relevant; one the classifier labels as the same aspect and subjective is partially relevant. We report P@5, Recall@50, MRR, NDCG@10 [11] and Success@10.

Results. Hybrid + RRF ranks best on MRR and NDCG@10 and ties BM25 on P@5 (Table 7 and Fig. 5). Dense retrieval alone is weakest: it ranks topically related but off-attribute sentences high; fusion keeps BM25's precision and adds paraphrases further down. The gain over BM25 is not significant with 30 queries (ΔNDCG@10 = +0.020, CI $[-0.036, +0.073]$). The cross-encoder *hurts* (ΔNDCG@10 = -0.066, CI $[-0.114, -0.019]$): trained on MS MARCO web questions, it prefers sentences that answer a question in general, such as a long comfort description, over short sentences that name the attribute. We therefore use it only to order candidates that already mention the requested sub-attribute. Success@10 is 1.0 everywhere, so differences are about ordering and coverage. These relevance labels are reproducible but favour explicit mentions, which advantages BM25 and under-counts dense paraphrases ("knees touching the seat in front"); human judgements would make the comparison stronger.

Which innovation matters? (E1b) E1 fixes the constraints and the query and varies the retriever. To measure the innovations of Table 4 themselves, we rebuild the 30 units from the first version upwards (Table 8). Relevance here does not reuse the expansion vocabulary: a hit counts only if it satisfies the question's constraints, checked from metadata, and the local Qwen3-4B, shown only the attribute's English name, says the sentence is about it (yes/no, 620 judgements). The judge answered 22 of 23 hand-written calibration sentences correctly; a spot check of 40 random judgements found 6 errors, mostly generic complaints about the airline accepted as relevant, so absolute precision is somewhat optimistic for every row.

Hard constraints are by far the largest gain. Without them only 16–35% of the top 10 come from the requested airline and cabin, because the sentence about legroom rarely names

Table 7: E1: retrieval quality on 30 airline × sub-attribute queries (automatic graded relevance). Best per column in bold; latency is the warm median per query including metadata filtering. These are the retriever configurations; the deployed pipeline also scans the filtered slice for sub-attribute mentions (Section 4.1).

Configuration	P@5	Recall@50	MRR	NDCG@10	Success@10	Latency
BM25	**0.880**	**0.403**	0.950	0.790	1.000	0.534 s
Dense (MiniLM)	0.753	0.307	0.867	0.686	1.000	**0.057 s**
Hybrid + RRF (default)	**0.880**	0.381	**0.961**	**0.810**	1.000	0.592 s
Hybrid + RRF + cross-encoder	0.827	0.387	0.933	0.744	1.000	0.714 s

Table 8: E1b: the retrieval innovations of Table 4 added one at a time on the 30 units of E1, all with hybrid retrieval and RRF. *In scope*: share of the top 10 that satisfies the question's airline, cabin and other constraints (exact, from metadata). P@5, MRR and NDCG@10 (binary) count a hit as relevant only if it is in scope and the LLM judge says it is about the attribute. *Direct*: share of the top 10 that names the sub-attribute explicitly (keyword match; shares vocabulary with R2). Δ: NDCG@10 change from the row above, with 95% bootstrap interval.

Configuration	In scope	P@5	MRR	NDCG@10	Direct	ΔNDCG@10 [95% CI]
First version: question as typed, whole corpus	0.213	0.053	0.102	0.057	0.000	
+ parsed into English keywords (cross-lingual)	0.350	0.160	0.389	0.173	0.070	+0.116 [+0.060, +0.175]
+ R1 hard metadata constraints	**1.000**	0.953	**1.000**	0.934	0.670	+0.761 [+0.673, +0.838]
+ R2 sub-attribute expansion (deployed)	**1.000**	**0.967**	0.983	**0.951**	**0.837**	+0.016 [−0.022, +0.057]
Deployed without R1	0.160	0.127	0.297	0.121	0.097	−0.830 [−0.892, −0.763]

the airline; with them every hit does, and NDCG@10 rises by 0.761. Parsing the Chinese question into English keywords lifts NDCG@10 from 0.057 to 0.173, since neither BM25 nor the English MiniLM can match Chinese text to an English corpus. Sub-attribute expansion adds little once constraints are in place (+0.016, not significant): in a slice of one airline and cabin, the attribute name alone already finds on-topic sentences. What it raises is the share of hits that name the sub-attribute explicitly (0.67 to 0.84), which the full-slice scan (R3) relies on. R3 itself needs no ranking metric: even the best configuration's top 50 holds only 38% of the direct-mention sentences (Recall@50, Table 7), so a verdict computed from the top k would see a minority of the evidence.

5.3 E2: How fast is a query? (Q2)

Table 9 reports five of the 20 evaluation questions, chosen to cover the supported query types, as warm runs with the local model on. *Retrieval* covers parsing, filtering, BM25, dense search, fusion, reranking and evidence selection; *first token* is measured from submission; *total* runs until the checked answer is shown. Figure 6 shows that retrieval is a small share of the wait: the 4B model on a laptop dominates, which we mitigate by streaming. Timed alone over the full pool without filters (the 30 E1 queries, warm medians), BM25 top-20 takes 503 ms, FAISS top-20 20 ms, hybrid retrieval (both top-100 lists and RRF) 519 ms and cross-encoder reranking of 100 candidates 108 ms (`experiments/final_auto_eval/component_latency.py`); scoring all sentences with BM25 dominates. With the LLM off, the program-written path averaged 5.0 s per question over a 40-question smoke test.

Table 9: E2: five representative queries (asked in Chinese; original wording and results in Appendix C). Answer: model draft (D), repaired draft (R) or program-written (P). The All-20 rows include the cold first question.

	Query	Retr.	1st tok.	Total	Ans.
Q05	Emirates economy: refunds and rebooking?	2.2 s	23.8 s	26.8 s	D
Q10	Qatar business: film selection?	2.2 s	18.4 s	33.6 s	R
Q24	Cathay vs. SIA economy legroom?	5.1 s	28.4 s	57.8 s	R
Q28	SIA economy: legroom, recline, comfort, cleanliness?	9.7 s	27.4 s	53.6 s	P
Q30	SIA economy: food, value, ground service?	6.9 s	50.8 s	82.0 s	P
All 20: median		2.9 s	24.3 s	40.8 s	
All 20: P95		10.5 s	41.9 s	72.3 s	

5.4 E3: How good is the classifier? (Q4)

Table 10 gives the summary metrics used for model selection (macro-F1 over aspects and polarities on the 919 subjective sentences, joint accuracy, recall of each subjectivity class); Table 11 gives per-class precision, recall and F1 as the brief requests, and Fig. 7 breaks aspect F1 down by class.

Results. Objective sentences are the main weakness of weak supervision: with 179 objective training examples, v4.1 finds 7.4% of them and the embedding model 21.0%; teacher labels raise this to 66.7% at a cost of 3 points of subjective recall, and three-class macro-F1 rises from 57.2 to 78.5. Aspect errors are concentrated: the student's most frequent confusions are *staff* predicted as *food* (18 sentences, mostly crew handling of the meal service, e.g. "No refills of drinks offered during or after the meal"), *ground* as

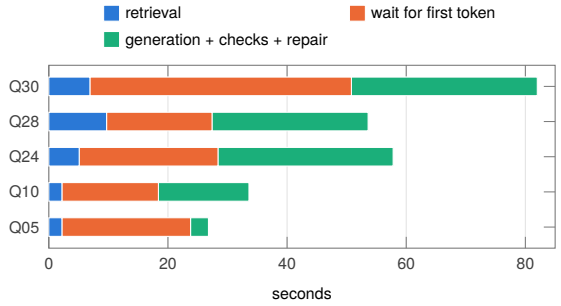


Figure 6: E2: where the time goes for the queries in Table 9.

Table 10: E3: classifier quality on the gold set (%). Aspect and polarity F1 are macro-averaged over 919 subjective sentences. Shaded: the distilled student, selected to replace v4.1 (Section 5.5). Joint: aspect and polarity both correct; S-rec./O-rec.: recall of subjective/objective sentences over all 1,000. v4.1 is the weak-supervision classifier the interface uses; r36 is the LLM teacher (Table 5); targets were fixed at the project start.

Model	Asp.	Pol.	Joint	S-rec.	O-rec.
v4.1: TF-IDF + rules (weak)	61.1	82.1	56.1	99.9	7.4
TF-IDF + MiniLM (weak)	68.1	85.2	61.5	**100.0**	21.0
Distilled (weak + teacher)	70.4	86.2	63.5	96.8	66.7
r36 LLM teacher	**74.1**	**91.8**	**70.8**	96.8	**76.5**
Project target	85	90	80	97	–

value (16, mostly refunds and compensation) and *overall* vs. *other*. These sentences genuinely concern two aspects, so part of the gap to the 85% target reflects the single-aspect label definition. No model, not even the LLM teacher, reaches every target; we report this rather than tune on the gold set.

5.5 E4: Can a fast classifier inherit the LLM's labels? (Q5)

Motivation. The label index drives the tally users see. Built with v4.1 it marked only 0.6% of the corpus objective, against 8.1% in gold, so almost every factual sentence was counted as an opinion. The teacher is far better but would need about three days for the corpus.

Protocol. (i) We drew 10,003 sentences from 1,154 parent reviews in the *development* split of the weak labels, excluding every gold sentence, its parent review and its normalised text, and any text seen in weak-label training. (ii) The teacher labelled all of them in 89 minutes with no invalid outputs (12.8% objective). (iii) We split by parent review into pseudo-train (7,355) and pseudo-dev (2,648). (iv) Before training we fixed three students with identical features that differ only in supervision: *weak* (weak labels only, reproducing a teammate's model), *teacher* (teacher labels only) and *mixed* (both, teacher examples up-weighted so each source carries equal weight). (v) The winner was selected by a fixed rank score against the *teacher* on pseudo-dev (0.35 aspect F1 $+\,0.20$ polarity F1 $+\,0.35$ joint $+\,0.10$ subjective recall, minus penalties when subjective recall falls below 97% or objective recall below 60%); a file-read guard prevented access to gold. (vi) Also fixed in advance: the winner replaces v4.1 only if it beats v4.1 on gold in both rank score and joint accuracy.

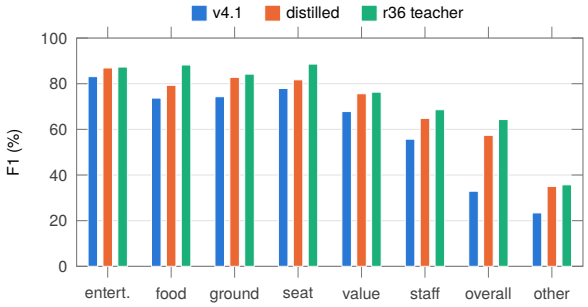


Figure 7: E3: aspect F1 per class on subjective gold sentences (supports: entertainment 115, food 99, ground 238, seat 86, value 130, staff 103, overall 101, other 47). Concrete aspects are easy; *overall* and *other* are hard for every model.

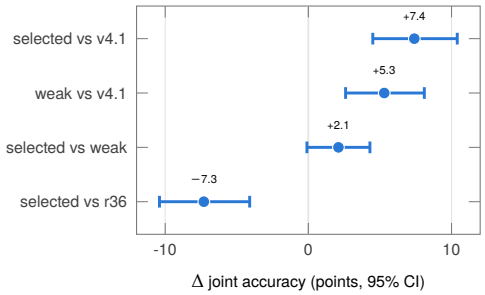


Figure 8: E4: paired differences in joint accuracy on gold (parent-review bootstrap). CIs: $[+4.5, +10.4]$, $[+2.6, +8.1]$, $[-0.1, +4.3]$, $[-10.4, -4.1]$.

Results. The mixed student was selected on pseudo-dev and passed the adoption rule on gold (Table 12). Replacing the v4.1 features by TF-IDF + MiniLM gives +5.3 points of joint accuracy (confounded by v4.1's rule layer); adding teacher labels gives a further +2.1, not significant at the 95% level (CI $[-0.1, +4.3]$, Fig. 8), and lifts objective recall from 21.0% to 66.7%. Teacher labels are valuable per example: the teacher-only student, trained on 7.4k sentences, has the best aspect F1 and objective recall of the three, but is weaker on polarity than models that also see the 231,779 weak labels, so mixing works best overall. Rebuilding the 529,031-label index takes 18 s instead of an estimated three days; the corpus's objective share goes from 0.6% to 13.0% (teacher: 12.8% on the sample), and 32% of aspect and 24% of polarity labels change. The student remains 7.3 points below its teacher.

Status. Because a third of the labels change, the interface switches to the new index only after the downstream statistics are re-run; E5 and E6 below were produced with the v4.1 labels, and E7 uses a separate diagnostic classifier (TF-IDF and logistic regression).

5.6 E5: Are the answers grounded? (Q4)

We evaluate the whole system on the 20 questions with RAGAS-style metrics [23]: *faithfulness* (share of answer

Table 11: E3: precision / recall / F1 (%) per class. The three-class label is the `label` column of eval.xls. Polarity is computed on the 919 gold-subjective sentences, where a wrong "objective" prediction counts as a miss.

	v4.1 (previous)			Distilled student			r36 teacher		
Class	P	R	F1	P	R	F1	P	R	F1
Three-class label									
negative	76.8	88.3	82.2	87.9	87.4	87.6	91.4	93.4	92.4
positive	76.6	74.9	75.7	82.0	82.2	82.1	92.0	86.5	89.2
objective	85.7	7.4	13.6	65.1	66.7	65.9	68.1	76.5	72.1
macro average	79.7	56.9	57.2	78.3	78.8	**78.5**	83.8	85.5	84.6
accuracy			76.8			**83.8**			89.5
Subjectivity detection									
subjective	92.4	99.9	96.0	97.1	96.8	96.9	97.9	96.8	97.4
objective	85.7	7.4	13.6	65.1	66.7	65.9	68.1	76.5	72.1
Polarity detection (gold-subjective sentences)									
negative	83.9	88.3	86.0	89.9	87.4	88.6	93.3	93.4	93.3
positive	81.5	74.9	78.1	85.4	82.2	83.8	94.1	86.5	90.2

Table 12: E4: supervision ablation. Left: agreement with the teacher on pseudo-dev (used for selection). Right: the same models on human gold, scored once after selection. Features are identical; only the training labels differ.

	vs. teacher, pseudo-dev			vs. human gold			
Supervision	**Joint**	**Obj. rec.**	**Rank**	**Aspect F1**	**Joint**	**Obj. rec.**	**Rank**
v4.1 production (reference)	54.0	3.9	0.637	61.1	56.1	7.4	0.648
Weak only	59.9	18.9	0.692	68.1	61.5	21.0	0.705
Teacher only (7.4k)	64.0	**75.7**	0.662	**71.4**	61.4	**76.5**	0.651
Weak + teacher (selected)	**65.8**	67.1	**0.729**	70.4	**63.5**	66.7	**0.735**

Table 13: E5: answer-level metrics of the guarded system (20 questions).

Metric	**Score**	**95% CI**
Faithfulness	0.843	[0.800, 0.890]
Answer relevance	0.988	[0.963, 1.000]
Context precision	0.600	[0.517, 0.681]

claims supported by evidence), *answer relevance* and *context precision* (share of displayed evidence relevant to the question). The judge is the local Qwen3-4B with a fixed rubric; it sees the displayed evidence and the program-computed ratings and tallies, so correct aggregate numbers are not marked unsupported.

Results. Answers stay on topic and are mostly grounded (Table 13). Context precision is the weakest: about 40% of displayed sentences were judged not directly relevant, partly because we deliberately show a counter-example and broader-aspect context (E6). The scores describe the *system*: the model's first draft was used for 1 question, its repair for 3 and the program-written answer for 16, because the 4B model's drafts often failed the strict citation or number checks. Since generator and judge are the same model, these are reproducible automatic proxies, not independent human scores.

5.7 E6: What does each answer-side component buy? (Q5)

The brief asks for the gain of each innovation alone and in combination. We therefore ran a full 2^3 factorial ablation of the three factors of Section 4.3: X (full-pool statistics), Y (balanced, label-consistent examples) and Z (validation, one repair, program-written fallback).

Protocol. We use 20 English single-airline, single-attribute questions from the development smoke set (Appendix D), fixed before scoring. Each question has a frozen pool of up to 30 candidates ranked without labels; all eight configurations share the pool, the prompt and a budget of four citations. With X off, corpus counts and the overall direction are hidden from generation, validation and fallback; with Y off, the top four candidates are used; with Z off, the model's draft is returned unchecked. Qwen3-4B-Instruct-2507 (BF16, Transformers backend on an AMD Radeon RX 7800 XT) both generates and judges, using the metrics of E5 with answer relevance scored 0–4 and rescaled. Each question gets one draft per X/Y setting, shared by its Z-off and Z-on variants, so the run produced 80 drafts, 54 repair attempts and 160 final answers. Intervals come from 2,000 paired bootstrap resamples over questions (seed 6127), without multiple-comparison adjustment.

Results. Y + Z is the most faithful configuration, 0.940, $+0.053$ over the baseline (CI $[+0.004, +0.106]$), and Z alone is also positive ($+0.030$, $[+0.003, +0.063]$) (Table 14 and Fig. 9). Y alone keeps answer relevance at 1.000 and

Table 14: E6: factorial ablation (20 questions per configuration). Code XYZ, 1 = component on. Δ: paired difference in faithfulness against the baseline, with 95% bootstrap interval.

Code	Configuration	Faithfulness	Δ vs. baseline [95% CI]	Answer rel.	Context prec.
000	Baseline	0.887	–	**1.000**	0.808
100	X only	0.860	−0.028 [−0.078, +0.024]	0.988	0.808
010	Y only	0.918	+0.031 [−0.029, +0.088]	**1.000**	**0.871**
001	Z only	0.917	+0.030 [+0.003, +0.063]	**1.000**	0.808
110	X + Y	0.902	+0.014 [−0.031, +0.061]	**1.000**	0.854
101	X + Z	0.868	−0.020 [−0.080, +0.039]	0.925	0.808
011	Y + Z	**0.940**	+0.053 [+0.004, +0.106]	0.975	**0.871**
111	X + Y + Z (deployed)	0.885	−0.002 [−0.074, +0.065]	0.975	0.854

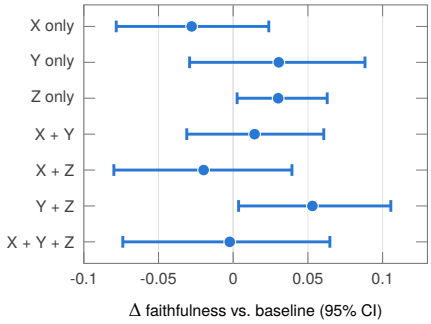


Figure 9: E6: faithfulness of each configuration relative to the baseline (data of Table 14). Only Z alone and Y + Z have intervals entirely above zero.

Table 15: E6: main and interaction effects (20 paired questions). A main effect is the on-minus-off difference averaged over the four settings of the other two factors; XY is the difference between X's effect with Y on and off, averaged over Z, and XYZ the difference between the XY interactions with Z on and off.

Effect	Faithfulness [95% CI]		Rel.	Ctx.
X	−0.037	[−0.075, +0.001]	−0.022	−0.008
Y	+0.028	[−0.002, +0.059]	+0.009	+0.054
Z	+0.011	[−0.015, +0.035]	−0.028	0.000
XY	+0.003	[−0.072, +0.070]	+0.044	−0.017
XZ	−0.031	[−0.076, +0.017]	−0.031	0.000
YZ	−0.016	[−0.069, +0.035]	+0.006	0.000
XYZ	−0.017	[−0.097, +0.060]	+0.063	0.000

raises context precision from 0.808 to 0.871 while using fewer citations (3.50 vs. 3.95 on average). The deployed configuration, X + Y + Z, is level with the baseline on faithfulness (−0.002) and X + Z lowers answer relevance the most (−0.075, [−0.125, −0.025]). Averaged over the other factors (Table 15), Y has the largest positive main effect, Z a small positive one and X a negative one; all three intervals include zero and the largest interaction is XZ. The gains are therefore not additive (all three together are level with the baseline), although with 20 questions no single interaction is significant. X is partly penalised by construction: it adds aggregate claims that the judge must verify, whereas an answer that only paraphrases quotations has little to get wrong. Faithfulness measures how the answer uses its context, not whether the overall opinion summary is right, which is what X is for; the targeted checks below measure that directly.

Table 16: E6: targeted checks, each component measured on the property it targets. The last two rows are sanity checks of the mechanism rather than measured gains.

Component (measured property)	Without	With
X: full-pool verdict vs. top-10 vote (direction agrees with all matching reviews)	12/18	**18/18**
Y: balanced evidence vs. naive order (top four show both sides when both exist)	11/17	**17/17**
Y: consistency filter (clear label–text conflicts among the examples shown for the 18 cases)	13	**0**
Z: direction guard vs. citation check only (wrong-direction answers rejected)	0/18	**18/18**

Cost. Z relies heavily on its fallback: of the 80 Z-on answers, 26 used the first draft, 7 the repair and 47 the program-written answer, and with X on every question triggered a repair. Median generation time is 9.78 s for X + Y + Z against 4.83 s for the baseline and 4.20 s for Y alone (shared retrieval and judging excluded).

Targeted checks. The factorial run scores final answers. Earlier checks (Table 16) measured each component on the failure it was designed to fix, over 18 airline × sub-attribute cases with a clear full-pool direction (17 with both positive and negative reviews); the baseline genuinely disables label-assisted ordering and balanced pre-selection. The direction-guard row is adversarial: we fed the validator 18 answers that cite real evidence but state the opposite direction; a citation check alone accepted all of them and the guard rejected all. The reference direction comes from the automatic labels, so 100% shows that the verdict no longer depends on which ten sentences are on screen, not that the labels match human judgement. A seven-question cumulative run with templated answers showed the same trade-off as the factorial: direction accuracy rose from 85.7% to 100% and context precision from 0.929 to 1.000, while judged faithfulness fell from 0.857 to 0.793, because a counter-example inside a mostly positive verdict was marked as not fully supported.

5.8 E7: Do text-derived rankings agree with passenger ratings? (Q5)

Sentence metrics measure local predictions and faithfulness measures grounding; neither shows whether opinion mined from text preserves the comparisons users make between airlines. We therefore rank airlines on economy seat expe-

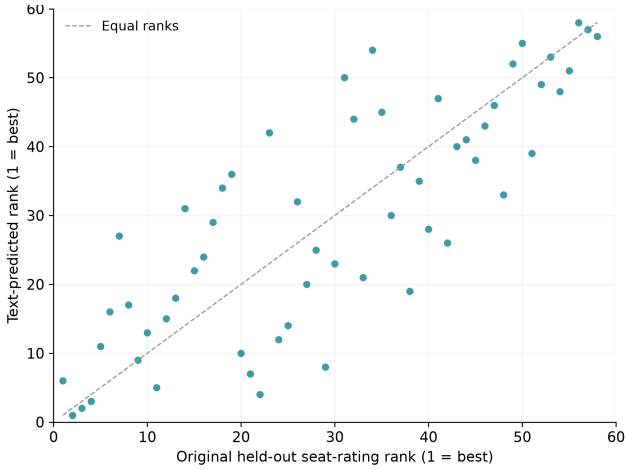


Figure 10: E7: text-predicted vs. held-out rating rank for 58 airlines (1 = best); the dashed line marks equal ranks. Spearman $\rho = 0.810$, 95% CI $[0.682, 0.838]$.

rience twice. The *text rank* uses a dedicated diagnostic classifier (TF-IDF 1–2-grams and logistic regression, retrained on weak labels without the held-out reviews) that finds subjective seat sentences and predicts their polarity; each review's share of positive sentences is averaged per airline, so every review has equal weight. The *reference rank* is the airline's mean `seat_comfort` rating over held-out reviews. About 20% of parent reviews were held out by a fixed hash of the review ID; training excluded them, their normalised texts and the reviews of the 1,100 labelled sentences (audited overlap zero), and the predictions were frozen and hashed before the held-out ratings were read. An airline needs at least 30 reference and 15 predictable reviews.

Results. 58 airlines qualify. The held-out reference set holds 8,365 reviews with a valid seat rating, 3,152 of them with a usable text prediction. The rankings agree closely: Spearman $\rho = 0.810$, 95% CI $[0.682, 0.838]$, from 2,000 paired resamples of parent reviews over the fixed airline set (Fig. 10). Opinion mined from text and aggregated per review therefore preserves clear differences between airlines, which supports using it for comparison questions. The rating is a reference for overall seat comfort, not a gold standard for each sub-attribute, and the interval does not include the uncertainty of retraining the classifier.

5.9 E8: Performance and scalability (Q4)

Retrieval and labelling scale well: indexes and labels are built offline, so query time depends on the filtered slice, not on corpus size; the student labels ≈30,000 sentences per second (Table 17). Generation is the bottleneck. Our single MLX server processes one request at a time, so two simultaneous users roughly double the wait; serving more users needs several model workers or a GPU server with a batched engine such as vLLM. The 5 s program-only path is a usable fallback under load.

Table 17: E8: performance on an Apple M1 Pro (16 GB). Query timings are medians (P95) over the 20 evaluation questions.

Measure	Value
Retrieval and evidence selection	2.9 s (10.5 s)
Submission to first streamed token	24.3 s (41.9 s)
Submission to final checked answer	40.8 s (72.3 s)
Program-only answer (LLM off)	5.0 s / query
Short LLM request (120 tokens), 1 or 2 at once	≈1 s each, serialised
Student labelling of 529,031 sentences	18 s
One-off MiniLM encoding of all sentences	573 s
LLM teacher labelling	≈0.5 s / sentence
KB rebuild / BM25 build / FAISS build	14 min / 6 s / 4.6 min
Cost per query	$0 (all local)

6 Discussion and Limitations

Automatic judges. Retrieval relevance and RAG metrics are automatic, and the RAG judge is the generator itself. An audit of the E6 judgements found 70 of 134 cached answer scorings listing more unsupported examples than the number of unsupported claims they counted, and reordering the same passages changed context precision for one question; bootstrap intervals do not capture such judge error. Human relevance judgements and blind human review of the answers, ideally with an independent judge model, would strengthen E1, E5 and E6.

Small generator and strict checks. 16 of 20 answers in E5 and 47 of 80 Z-on answers in E6 used the program-written fallback; in Q01 of E6 a repair fixed a wrong number but changed the required wording of the statistics sentence and was rejected. Separating semantic errors from format violations, and measuring the false-rejection rate, is the next step; a larger local model or a short fine-tune on answers that passed the checks would also help.

Classifier below target. Aspect F1 is 70.4 against a target of 85. Since teacher labels proved more valuable per example than weak labels, the cheapest next steps are more teacher labels (30–50k sentences, a few GPU hours), fine-tuning a small encoder on them, and a cascade that sends only low-confidence sentences to the LLM. Allowing more than one aspect per sentence would also match the data better.

Airline names in English. Late testing found "…Cathay Pacific or Singapore Airlines?" parsed as a single-airline question, because the corpus spells the name "Cathay Pacific Airways". A table of common English short names now resolves this (it changes the parse of none of the evaluation questions); names outside the table still need the corpus spelling, a Chinese name or the sidebar.

Data age. Reviews end in mid-2019. The interface states this on every page; answers describe historical experiences, not current service.

7 Conclusion

We built a local RAG system for airline-service questions that keeps constraints, works at the level of fine-grained attributes and gives verdicts a user can verify. Four lessons stand out. First, structured constraints and attribute routing fixed the failing queries of Table 4: hard metadata constraints alone raise NDCG@10 by 0.76 (E1b), far more than any change of ranking model, and a general-purpose cross-encoder even hurt. Second, distilling an LLM labeller into a linear classifier recovers most of its gain in objective detection and about half of its gain in joint accuracy at a tiny fraction of its cost, provided selection is kept away from the test set. Third, balanced evidence plus validation with a program-written fallback gives the most faithful final answers (Y + Z is the best of eight configurations), but the gains are not additive: exact full-pool statistics fix the verdict while lowering judged faithfulness. Finally, airline rankings mined from text agree with held-out passenger ratings ($\rho = 0.81$), which supports answering comparison questions from review text.

Data and Code Availability

Release page: `https://github.com/ntu-ai6127-g-13/ai6127-airline-rag/releases/tag/v1.0-submission`

Data archive: `https://github.com/ntu-ai6127-g-13/ai6127-airline-rag/releases/download/v1.0-submission/ai6127-g13-data.zip`
`chunks.parquet` (knowledge base), `eval.xls` and the frozen gold CSV, the 20 evaluation questions with retrieved evidence and generated answers, the 30-query retrieval benchmark with per-query results, label indexes (v4.1 and distilled), classifier predictions on gold, teacher labels, the ablation case files, the factorial-ablation run (protocol, prompts, 160 answers, judge scores) and the airline-ranking predictions.

Code archive: `https://github.com/ntu-ai6127-g-13/ai6127-airline-rag/archive/refs/tags/v1.0-submission.zip`
`src/`, `app/`, `scripts/`, `experiments/`, `tests/`, `requirements.txt` and a README with build and run instructions.

Contribution Statement

All members contributed to the project. As recommended in the brief, we request a single grade shared by the whole group.

References

[1] Stephen Robertson and Hugo Zaragoza. "The Probabilistic Relevance Framework: BM25 and Beyond". In: *Foundations and Trends in Information Retrieval* 3.4 (2009), pp. 333–389.

[2] Nils Reimers and Iryna Gurevych. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: *Proceedings of EMNLP-IJCNLP*. 2019, pp. 3982–3992.

[3] Wenhui Wang et al. "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers". In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020.

[4] Jeff Johnson, Matthijs Douze, and Hervé Jégou. "Billion-scale similarity search with GPUs". In: *IEEE Transactions on Big Data* 7.3 (2021), pp. 535–547.

[5] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. "Reciprocal Rank Fusion Outperforms Condorcet and Individual Rank Learning Methods". In: *Proceedings of SIGIR*. 2009, pp. 758–759.

[6] Payal Bajaj et al. *MS MARCO: A Human Generated MAchine Reading COmprehension Dataset*. 2016. arXiv: 1611.09268.

[7] Rodrigo Nogueira and Kyunghyun Cho. *Passage Reranking with BERT*. 2019. arXiv: 1901.04085.

[8] Nandan Thakur et al. "BEIR: A Heterogeneous Benchmark for Zero-shot Evaluation of Information Retrieval Models". In: *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*. 2021.

[9] Shitao Xiao et al. *C-Pack: Packaged Resources To Advance General Chinese Embedding*. 2023. arXiv: 2309.07597.

[10] Luyu Gao et al. "Precise Zero-Shot Dense Retrieval without Relevance Labels". In: *Proceedings of ACL*. 2023.

[11] Kalervo Järvelin and Jaana Kekäläinen. "Cumulated Gain-Based Evaluation of IR Techniques". In: *ACM Transactions on Information Systems* 20.4 (2002), pp. 422–446.

[12] Maria Pontiki et al. "SemEval-2014 Task 4: Aspect Based Sentiment Analysis". In: *Proceedings of the 8th International Workshop on Semantic Evaluation*. 2014, pp. 27–35.

[13] Wenxuan Zhang et al. "A Survey on Aspect-Based Sentiment Analysis: Tasks, Methods, and Challenges". In: *IEEE Transactions on Knowledge and Data Engineering* 35.11 (2023).

[14] Erik Cambria et al. "SenticNet 7: A Commonsense-based Neurosymbolic AI Framework for Explainable Sentiment Analysis". In: *Proceedings of LREC*. 2022, pp. 3829–3839.

[15] Hu Xu et al. "BERT Post-Training for Review Reading Comprehension and Aspect-based Sentiment Analysis". In: *Proceedings of NAACL-HLT*. 2019, pp. 2324–2335.

[16] Kevin Scaria et al. "InstructABSA: Instruction Learning for Aspect Based Sentiment Analysis". In: *Proceedings of NAACL (Short Papers)*. 2024.

[17] Wenxuan Zhang et al. "Sentiment Analysis in the Era of Large Language Models: A Reality Check". In: *Findings of NAACL*. 2024.

[18] Fabrizio Gilardi, Meysam Alizadeh, and Maël Kubli. "ChatGPT Outperforms Crowd Workers for Text-Annotation Tasks". In: *Proceedings of the National Academy of Sciences* 120.30 (2023), e2305016120.

[19] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. *Distilling the Knowledge in a Neural Network*. 2015. arXiv: 1503.02531.

[20] Wenpeng Yin, Jamaal Hay, and Dan Roth. "Benchmarking Zero-shot Text Classification: Datasets, Evaluation and Entailment Approach". In: *Proceedings of EMNLP-IJCNLP*. 2019.

[21] Moritz Laurer et al. “Less Annotating, More Classifying: Addressing the Data Scarcity Issue of Supervised Machine Learning with Deep Transfer Learning and BERT-NLI”. In: *Political Analysis* 32.1 (2024), pp. 84–100.

[22] Patrick Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems*. Vol. 33. 2020.

[23] Shahul Es et al. “RAGAS: Automated Evaluation of Retrieval Augmented Generation”. In: *Proceedings of EACL: System Demonstrations*. 2024, pp. 150–158.

[24] Tianyu Gao et al. “Enabling Large Language Models to Generate Text with Citations”. In: *Proceedings of EMNLP*. 2023.

[25] Akari Asai et al. “Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection”. In: *International Conference on Learning Representations*. 2024.

[26] efehandanisman. *Skytrax Airline Reviews*. Kaggle dataset. URL: https://www.kaggle.com/datasets/efehandanisman/skytrax-airline-reviews (visited on 2026-09-07).

[27] Robyn Speer. *ftfy: fixes text for you*. Python library. URL: https://github.com/rspeer/python-ftfy.

[28] Matthew Honnibal et al. *spaCy: Industrial-strength Natural Language Processing in Python*. 2020. DOI: 10.5281/zenodo.1212303.

[29] Qwen Team. *Qwen3 Technical Report*. 2025. arXiv: 2505.09388.

[30] Joseph L. Fleiss. “Measuring Nominal Scale Agreement Among Many Raters”. In: *Psychological Bulletin* 76.5 (1971), pp. 378–382.

A Reproducing the Results

All commands run from the repository root with Python 3.11. Experiment scripts never overwrite earlier results: most refuse an existing output directory, and the long-running ones resume from their checkpoints; the distillation scripts also record SHA-256 hashes of their inputs.

```
# environment and data (Q1)
python3.11 -m venv .venv && .venv/bin/pip install -r requirements.txt
.venv/bin/python -m spacy download en_core_web_sm
.venv/bin/python scripts/download_data.py
.venv/bin/python -m src.build_kb                        # about 14 min

# indexes
.venv/bin/python -m src.retriever build-bm25
.venv/bin/python -m src.retriever build-dense --batch-size 64
.venv/bin/python -m src.label_index --model outputs/weak_rebuild_20261019/model.joblib \
    --output-dir outputs/label_index_replay     # the archive already has label_index_20261019

# interface (Q2): local model server, then the app
.venv/bin/python scripts/setup_query_model.py           # once: pinned Qwen3-4B weights
outputs/query_quality_runtime/venv311/bin/python scripts/local_qa_server.py
.venv/bin/python scripts/run_query_app.py --port 8517

# retrieval benchmark and targeted checks (E1, E6); RAG metrics and latency (E2, E5)
.venv/bin/python -u experiments/final_auto_eval/run.py --output-dir outputs/final_auto_eval_replay
.venv/bin/python -u experiments/final_rag_eval/run.py --output-dir outputs/final_rag_eval_replay
# retrieval innovation ablation (E1b); the LLM judge needs the model server running
.venv/bin/python -u experiments/final_auto_eval/retrieval_ablation.py \
    --output-dir outputs/retrieval_ablation_replay

# factorial ablation (E6) and airline ranking (E7): Linux GPU, Transformers backend;
# full commands in experiments/downstream_evaluation/README.md and FACTORIAL.md
.venv/bin/python scripts/local_qa_server.py --backend transformers --device cuda
P=outputs/downstream_protocol_replay
.venv/bin/python -m experiments.downstream_evaluation.freeze --output-dir $P
.venv/bin/python -m experiments.downstream_evaluation.ranking \
    --protocol $P/protocol.json --output-dir outputs/downstream_ranking_replay

# distillation (E3, E4); encode.py and evaluate_gold.py default to --device mps (use cpu/cuda
# elsewhere). The teacher step needs the MLX runtime and earlier r36 outputs, so the sample and
# its teacher labels ship in the data archive ($S); sample.py and label_teacher.py remake them.
D=experiments/distilled_label_index; O=outputs
S=$O/distill_sample_20261029; E=$O/distill_embeddings_20261029; R=$O/distill_students_replay
.venv/bin/python $D/encode.py --output-dir $E                                   # 5-10 min
.venv/bin/python $D/train_students.py --sample-dir $S --embeddings $E --output-dir $R
.venv/bin/python $D/evaluate_gold.py --selection $R/selection.json --output-dir $R/gold
.venv/bin/python $D/build_index.py --audit $R/gold/gold_audit.json \
    --selection $R/selection.json --embeddings $E --output-dir $O/label_index_distilled_replay
```

B Evaluation File Format

`eval.xls` (SHA-256 `fe0bcf0c···a477045`) is an Excel 97–2003 workbook written with `xlwt` from the frozen gold CSV: one sheet, 1,000 rows, no duplicate texts, and exactly the two benchmark columns `text` and `label`. The gold CSV (`data/eval/gold_human_frozen_1000_20261019_v1.csv`) has the same rows in the same order and adds the step-wise labels and provenance:

Column	Values	Meaning
`text`	string	The sentence, exactly as shown to annotators.
`label`	objective / positive / negative	Three-class sentiment label (benchmark format).
`subjectivity`	objective / subjective	Step 1 of the task.
`polarity`	positive / negative / NA	Step 2; `NA` for objective sentences (read with `keep_default_na=False`).
`aspect`	seat, staff, food, entertainment, ground, value, overall, other	Aspect the sentence is about.
IDs and provenance	sentence, review and chunk IDs; review status	Joins back to the knowledge base; audit trail of the human review.

Question sets. `data/eval/query_smoke_40_20261019.json` is a JSON array of 40 questions, each with `id`, `language`, `question`, the expected `aspect`/`aspects`, the expected `subtopics` (sub-attributes) and a

`comparison` flag. E1 parses Q01–Q25 into the 30 airline × sub-attribute units; E2 and E5 use Q01–Q12, Q21–Q25 and Q28–Q30. Results are JSONL with one record per question (`case`, `parsed`, `result` with the retrieved evidence and timings, `answer` with text, mode and citations, and `judge`), or CSV with one row per query and method.

C The Queries of Table 9: Wording and Results

Counts are reviews that directly match the attribute; percentages are over those with a clear direction. Full answers, with citations and evidence, are in `outputs/final_rag_eval_20261019/answers.jsonl`.

ID	Question as asked	Result
Q05	阿联酋航空经济舱的退款和改签处理怎么样?	Negative: 19 of 19 directional reviews (24 matches) negative; cites unpaid refunds after repeated contact and cancellation and seat fees.
Q10	卡塔尔航空商务舱的电影选择和娱乐内容怎么样?	Mixed: 15 positive, 15 negative (86 matches); a wide film selection versus dated content and poor TV choice.
Q24	国泰航空和新加坡航空经济舱腿部空间哪家评价更好?	No clear winner on legroom: Cathay Pacific 72.5% positive (50/69), Singapore Airlines 79.6% (78/98); Singapore's seat rating is higher (3.83 vs. 3.46 of 5).
Q28	新加坡航空经济舱的腿部空间、靠背调节、座椅舒适度和座椅清洁度分别怎么样?	Legroom positive (78/98, 79.6%); recline mixed (14/26); seat comfort positive (97/123, 78.9%); cleanliness too few to decide (7/10).
Q30	新加坡航空经济舱的餐食、性价比和地面服务分别怎么样?	Food 65.5% positive (289/441, rating 3.88); value 34.6% positive (75/217, rating 3.88); ground service 43.6% positive (163/374, rating 3.99).

D Questions of the Factorial Ablation

Each question reads "What do historical reviews say about [attribute] on [airline] in [cabin]?". These IDs are local to E6 and unrelated to the IDs of Table 9.

ID	Airline	Cabin	Attribute	ID	Airline	Cabin	Attribute
Q01	Singapore Airlines	Economy	legroom	Q11	British Airways	Prem. Econ.	recline / lie-flat
Q02	Air Canada rouge	Economy	legroom	Q12	Cathay Pacific	Business	staff attitude
Q03	Spirit Airlines	Economy	baggage	Q13	United Airlines	Economy	seat width
Q04	Alaska Airlines	Economy	delay handling	Q14	Turkish Airlines	Economy	food quality
Q05	Emirates	Economy	refunds / changes	Q15	AirAsia	Economy	customer support
Q06	Singapore Airlines	Economy	meal variety	Q16	Etihad Airways	Business	recline / lie-flat
Q07	Singapore Airlines	Economy	portion size	Q17	Frontier Airlines	Economy	baggage
Q08	Singapore Airlines	Economy	seat cleanliness	Q18	American Airlines	Economy	check-in
Q09	Emirates	Economy	Wi-Fi	Q19	Lufthansa	Economy	punctuality
Q10	Qatar Airways	Business	entertainment	Q20	Qantas Airways	Economy	drinks

E Aspect and Sub-attribute Taxonomy

Aspect	Sub-attributes (46 in total)
Seat	legroom, width, recline and lie-flat, comfort, condition, seat cleanliness, power, privacy, storage
Food	specific dishes, quality and taste, portion size, variety, drinks, special meals, temperature, meal-service process
Entertainment	content, screen, Wi-Fi, headphones and audio, system usability
Staff	attitude, efficiency, professionalism, availability, communication and language
Ground	punctuality, delay and cancellation handling, baggage, check-in, gate and boarding, lounge, connections, special assistance
Value	ticket price, refunds and changes, customer support, booking system, additional fees
Other	cabin cleanliness, lavatory, cabin temperature, amenities, cabin noise, aircraft and cabin condition
Overall	whole-trip judgements; no sub-attributes